

Case study

AquaPoly 511 for SB Coil Coatings

- **Background:** Multi-national customer needed PTFE-free replacement for AquaPoly 411.
- **System:** Solvent-based polyester crosslinked with melamine-formaldehyde resin.
- **Process:** Metal rolls are roll coated and heat cured at 400-500°F for 30-45 seconds followed by water quench.
- **Key properties / Function of the wax:**
 - Improved surface slip
 - Protection during post forming
 - Must maintain flexibility
 - Scratch and abrasion resistance
 - Reduced water spotting (metal roofs)
 - Easy dispersibility



Samples Tested

1. **MPP-611AL** – HDPE/AL
2. **Polyglide 1226XF** – HDPE/Ceramic
3. **MicroKlear 116AL** – HDPE/T1/AL
4. **MP-28AL** – FT/AL



Results

- The good: Scratch, abrasion, flexibility, gloss retention.
- The bad: Water spotting, Dispersibility/intermediate stability.

Development of AquaPoly 511

- Modified Oxidized PE base
- 6.0-8.0 um mean, 22.0 um max
- 117-123°C melt point
- Density: 0.97 g/cc

Results:

- Excellent water spotting resistance
- Improved dispersibility and stability in SB intermediate (Xylene, Toluene, n-Butanol)

